

Course Schedule

Course Schedule

Course Schedule

Week	Topic & Agenda Summary	Course Materials	Assignments & Due Dates
Wk. 1	- Principles of Making Justifiable Inferences - Design-Based Inferences and Randomization Tests	- Module 1, Parts 1 and 2 - Recommended: Ch. 1 in Statistical Sleuth - Lecture videos m1-s1 through m1-s12 - Install R, use the included R manual to run simple commands - Complete Lab 1 and Lab 2	Assignment #1 due Tuesday, Week 2.
Wk. 2	Inference for Comparing Two Means: Frequentist and Bayesian Approaches	- Module 2, Parts 1 and 2 - Recommended: Ch. 2 in Statistical Sleuth. - Lecture videos m2-s1 through m2-s11 - Complete Lab 3 - Complete Homework 2	
Wk.3	Statistical Inference for Nonparametric and Categorical Data	Module 2, Part 3 - Recommended: Ch. 2 and Ch. 18 in Statistical Sleuth. - Lecture videos m2-s12 through m2-s15 - Complete Lab 4 - Complete Homework	

Week	Topic & Agenda Summary	Course Materials	Assignments & Due Dates
		3 • Check Homework 2 solutions • Module 3, Part 1 • Recommended: Ch. 3 and Ch. 18 in Statistical Sleuth. • Lecture videos m3-s1 through m3-s7 • Complete Homework 4	
Wk.4	Transformations and Two-Sample Analysis	• Module 3, Part 2 • Recommended: Ch. 3 and Ch. 18 in Statistical Sleuth. • Lecture videos m3-s8 through m3-s11 • Complete Lab 5 • Complete Homework 5 • Check Homework 3 and 4 solutions	• Milestone #1 due Monday, Week 4, June 8
Wk.5	Analysis of Variance (ANOVA) for Multiple Group Comparisons	• Module 4, Part 1 • Recommended: Ch. 5 and Ch. 6 in Statistical Sleuth. • Lecture videos m4-s1 through m4-s8 • Complete Homework 6 • Check Homework 5 solutions	Exam 1, June 15 and June 16
Wk.6	• Post-Hoc Inference for Multiple Groups Using Linear Contrasts	• Module 4, Part 2 • Recommended: Ch. 5 and Ch. 6 in Statistical Sleuth. • Lecture videos m4-s9	

Week	Topic & Agenda Summary	Course Materials	Assignments & Due Dates
	Multiple Comparisons and Error Rate Control	through m4-s15 - Complete Lab 6 Module 4, Part 3 - Recommended: Ch. 5 and Ch. 6 in Statistical Sleuth. - Lecture videos m4-s16 through m4-s21 - Complete Lab 7 - Homework 8 - Check Homework 6 solutions	
Wk.7	Simple Linear Regression: Estimation, Inference, and Prediction	- Module 5, Parts 1 and 2 - Recommended: Ch. 7 and Ch. 8 in Statistical Sleuth. - Lecture videos m5-s1 through m5-s15 - Complete Lab 8 - Homework 9 - Check Homework 8 solutions	
Wk.8	Likelihood-Based Regression for Discrete and Count Data	- Module 5, Part 3 - Recommended: Ch. 7 and Ch. 8 in Statistical Sleuth. - Lecture videos m5-s16 through m5-s23 - Complete Lab 9 - Check Homework 9 solutions	Exam 2, July 7 and July 8

Week	Topic & Agenda Summary	Course Materials	Assignments & Due Dates
Wk.9	Multiple Linear Regression: Estimation, Inference, and Model Comparison	• Module 6, Part 1 • Recommended: Ch. 9 through Ch. 13 in Statistical Sleuth. • Lecture videos m6-s1 through m6-s7 • Complete Lab 10 • Homework 10	• Milestone #2 due Monday, Week 9, July 13.
Wk.10	Advanced Regression Models and Model Selection	• Module 6, Part 2 • Recommended: Ch. 9 through Ch. 13 in Statistical Sleuth. • Lecture videos m6-s8 through m6-s19 • Complete Lab 11 • Check Homework 10 solutions	
Wk.11	Model Diagnostics, Validation, and Case Studies	• Module 6, Part 3 • Recommended: Ch. 9 through Ch. 13 in Statistical Sleuth. • Lecture videos m6-s20 through m6-s28 • Complete Lab 12	
Wk.12	Finals Week		• Exam 3 week 12 • Final Project due Week 12 • Peer Evaluation due Week 12